When do local repairs of a P_3 packing combine into a cluster editing?

(working title; draft skeleton)

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Abstract

Li, Pilipczuk and Sorge study Cluster Editing parameterized by the excess ℓ of the solution size over a modification-disjoint packing H of induced P_3's. We ask which interaction structure of H makes this parameterization tractable, and use the treewidth w of the intersection graph $I(H)$ of the packing as the measure. We prove: (i) for $\ell = 0$ the problem is a CSP with domain size 3 whose constraint scopes are cliques of $I(H)$, hence solvable in $3^{w+1}n^{O(1)}$ given a tree decomposition; (ii) for every ℓ it is solvable in $7^{w+1}n^{2\ell+O(1)}$ given a tree decomposition (XP in ℓ, FPT in w), and domain size 7 cannot be lowered in the natural encoding; (iii) under ETH there is no $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$ algorithm even for $\ell = 0$ (exponent only, no base); (iv) w alone does not suffice (NP-hard for $w = 0$ with unbounded ℓ), and $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ are incomparable, with a tradeoff $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$ on stars. The main question, whether the problem is fixed-parameter tractable in $\ell + w$, is *open*; we do not prove or disprove it. Further statements obtained by our team are listed separately as unaudited claims, and are not part of the results above.

1 Introduction

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph G into a disjoint union of cliques. Every induced P_3 must be hit by an edit, so a packing H of induced P_3's that pairwise share no vertex pair (a *modification-disjoint* packing) gives the lower bound $|H|$ on the solution size. Li, Pilipczuk and Sorge [6] parameterize by the *excess* ℓ over this bound (problem CEaMP): is there a solution of size at most $|H|+\ell$? They show NP-hardness already for $\ell = 0$, with every vertex in at most 23 packed P_3's, and an $n^{2\ell+O(1)}$ algorithm when every vertex lies in at most two packed P_3's; whether the latter case is fixed-parameter tractable is open [6]. For vertex-disjoint packings the problem is FPT in ℓ [9].

The question of this paper is which measurable *interaction structure* of H makes the local repair choices (one edit per packed P_3, in three ways) combine into a global solution in $f(\ell, w)\, n^{O(1)}$ time. We take $w = \mathrm{tw}(I(H))$, where the intersection graph $I(H)$ has the packed P_3's as nodes and joins two of them when they share a vertex.

What is proved. For $\ell = 0$ the problem is a constraint satisfaction problem (CSP) with one variable of domain size 3 per packed P_3 and constraint scopes that are cliques of $I(H)$ (Section 3); dynamic programming over a tree decomposition gives $3^{w+1}n^{O(1)}$. For general ℓ, guessing the at most ℓ edits outside all packed P_3's reduces to the same CSP with domain size 7, giving $7^{w+1}n^{2\ell+O(1)}$ (Section 4). Under ETH the exponent $\Theta(w)$ cannot be improved to $o(w)$ even at $\ell = 0$; w alone is

insufficient; $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ are incomparable (Section 5). Section 6 collects structural lemmas on edits outside packed P_3's and a counterexample to a natural locality claim.

What is not proved. Whether CEaMP is FPT in $\ell + \mathrm{tw}(I(H))$ (or W[1]-hard) is open. The algorithm of Section 4 is XP in ℓ, and no hardness is known. Section 7 lists further statements that have not yet been audited and must not be cited as results; Section 8 states the open problems.

2 Preliminaries

Graphs are simple and undirected, $n = |V(G)|$. A *pair* is a 2-subset of $V(G)$; for a set S of pairs, $G \triangle S$ flips the adjacency of each pair in S. A graph is a *cluster graph* iff it has no induced P_3, i.e. no vertex triple with exactly two edges. A *solution* is a set S of pairs with $G \triangle S$ a cluster graph; $\mathrm{OPT}(G)$ is the minimum size of a solution.

A *packing* $H = \{P_1, \ldots, P_p\}$ is a set of induced P_3's $P_i = (a_i, b_i, c_i)$ ($a_ib_i, b_ic_i \in E$, $a_ic_i \notin E$). Write $\mathrm{pairs}(P_i)$ for the three pairs inside $V(P_i)$. H is *modification-disjoint* if the sets $\mathrm{pairs}(P_i)$ are pairwise disjoint; this is equivalent to: any two members share at most one vertex (two shared vertices would be a shared pair, and any two vertices of a P_3 form one of its pairs). All packings below are modification-disjoint. A pair is *packed* if it lies in $\bigcup_i \mathrm{pairs}(P_i)$ (its *owner* is then unique) and *unpacked* otherwise. For a vertex u, $Q(u)$ is the set of z with uz packed.

Problem CEaMP. Given G, a modification-disjoint packing H and $\ell \geq 0$, decide whether there is a solution S with $|S| \leq |H| + \ell$ (this is the convention of [6]; ℓ is the excess budget).

The *intersection graph* $I(H)$ has node set H and an edge PP' iff $V(P) \cap V(P') \neq \emptyset$. The P_3's through one vertex form a clique of $I(H)$. Hence (Helly property of tree decompositions) they lie in a common bag, and $c := \max_u |\{P \ni u\}| \leq \mathrm{tw}(I(H)) + 1$ and $|Q(u)| \leq 2c$. We write $w = \mathrm{tw}(I(H))$ with the convention $w = 0$ if $I(H)$ has no edge. A tree decomposition of width w is assumed *given* in the algorithmic statements; computing one costs $2^{O(w^3)}n$ exactly (Bodlaender) or $2^{O(w)}n$ for a factor-2 approximation (Korhonen), which would change the bases below (e.g. 7^{w+1} to 7^{2w+2}).

Lemma 1 (Solutions decompose)**.** *Let S be any set of pairs such that $G \triangle S$ is a cluster graph, $x_i = S \cap \mathrm{pairs}(P_i)$ and $U = S \setminus \bigcup_i \mathrm{pairs}(P_i)$. Then every $x_i \neq \emptyset$, $|S| = \sum_i |x_i| + |U|$, and $|U| \leq |S| - |H|$.*

Proof. If $x_i = \emptyset$ the triple $V(P_i)$ keeps its adjacencies in $G \triangle S$ and is an induced P_3. The pair sets are disjoint, so $|S| = \sum |x_i| + |U| \geq |H| + |U|$. □

Remark 2 (Maximality). If $H' \supseteq H$ adds t further pair-disjoint induced P_3's, then $|H'| + (\ell - t) = |H| + \ell$, so (G, H, ℓ) and $(G, H', \ell - t)$ ask the same question (a NO instance if $\ell < t$, since $\mathrm{OPT} \geq |H'|$). So for structure theory H may be taken *maximal* (no induced P_3 has all three pairs unpacked). Caution: $\mathrm{tw}(I(H'))$ may exceed $\mathrm{tw}(I(H))$.

3 The case $\ell = 0$ is a CSP on the intersection graph

Observation 3. *Let S be a solution with $|S| \leq |H|$. Then S has exactly one pair in each $\mathrm{pairs}(P_i)$ and no other pair, and $G \triangle S$ is a cluster graph iff no vertex triple is an induced P_3 of $G \triangle S$. Consequently $(G, H, 0)$ is a YES instance iff the following CSP is satisfiable: variables $x_i \in \{1, 2, 3\}$ (which pair of P_i is flipped), and for every vertex triple t the constraint "t is not an induced P_3 after flipping", whose truth depends on the owners' variables of the packed pairs of t only (unpacked pairs keep their status in G).*

Proof. By Lemma 1, $|x_i| \geq 1$ for all i and $|S| = \sum |x_i| + |U| \leq |H|$ force $|x_i| = 1$ and $U = \emptyset$. The rest is the definition of a cluster graph. □

The CSP has $\Theta(n^3)$ constraints, not $O(|H|)$.

Observation 4. *The set of owners of the packed pairs of any triple is a clique of $I(H)$ (at most three nodes). Hence the primal graph of the CSP is a subgraph of $I(H)$ (not every edge of $I(H)$ is witnessed by a triple), and a tree decomposition of $I(H)$ is one of the primal graph.*

Proof. Two pairs of one triple share a vertex v; their owners both contain v, so they coincide or are adjacent in $I(H)$. □

Theorem 5. *Given (G, H) and a tree decomposition of $I(H)$ of width w, CEaMP with $\ell = 0$ is solvable in $3^{w+1}n^{O(1)}$ time.*

Theorem 5 is the case $\ell = 0$ of Theorem 6 below (with a larger base there, 7, because for $\ell > 0$ a packed P_3 may receive several edits); its proof is the same dynamic program with domain $\{1, 2, 3\}$. Observation 4 is what makes the interaction of disjoint P_3's at $\ell = 0$ completely visible in $I(H)$: no transitivity-mediated interaction exists beyond shared vertices.

4 Excess edits: an XP-in-ℓ, FPT-in-w algorithm

Theorem 6. *Given (G, H, ℓ) and a tree decomposition of $I(H)$ of width w, CEaMP is solvable in $7^{w+1}n^{2\ell+O(1)}$ time.*

Proof. Let $\mathcal{D}_i$ be the 7 non-empty subsets of pairs(P_i). By Lemma 1, a solution of size $\leq |H| + \ell$ has the form $S = U \cup \bigcup x_i$ with $x_i \in \mathcal{D}_i$ and U a set of at most ℓ unpacked pairs; there are at most $\sum_{j \leq \ell} \binom{n^2/2}{j} \leq n^{2\ell}$ candidates for U.

Fix U and put $G_U = G \triangle U$. Since U avoids packed pairs, every P_i is still an induced P_3 of G_U with the same pairs, so H is still a modification-disjoint packing. For a triple t, the status of each unpacked pair of t in $G_U \triangle x$ is fixed, and each packed pair is determined by G_U and the variable of its owner. Hence t gives a constraint C_t ("t is not an induced P_3 in $G_U \triangle x$") whose scope is the owner set of t, a clique of $I(H)$ by Observation 4 (the proof there does not use G itself). Triples with empty scope are tested directly, in $O(n^3)$ time. Seek $x \in \prod \mathcal{D}_i$ satisfying all C_t and minimizing the unary cost $\sum_i |x_i|$.

Let $(T, \{B_s\})$ be the tree decomposition. By the Helly property each clique lies in a bag; assign each constraint to one such bag. For a node s and an assignment α of B_s, let $f_s(\alpha)$ be the minimum cost of $\sum |x_i|$ over variables in the subtree of s, among assignments extending α and satisfying all constraints assigned in the subtree. The standard recurrences (introduce, forget, join; see [3]) compute all f_s in time $7^{w+1}n^{O(1)}$ since there are $O(n^3)$ constraints and each table has $\leq 7^{w+1}$ entries. Accept iff for some U the minimum m_U satisfies $m_U + |U| \leq |H| + \ell$. Soundness: any feasible (U, x) gives the solution $U \cup \bigcup x_i$. Completeness: Lemma 1. □

Remark 7 (Why domain size 7). The empty value is never feasible, so 7 (not 8) suffices. It cannot be lowered to 3: an edited packed P_3 can receive two or three edits in every optimum. For two: $n = 5$, $E = \{02, 04, 12, 23, 24\}$, $H = \{(1, 2, 3)\}$, OPT $= 2$, unique optimum $\{12, 23\}$. For three: $n = 6$, $E = \{01, 02, 03, 04, 12, 14, 15, 24, 25, 35, 45\}$, $H = \{(0, 3, 5)\}$, OPT $= 3$, unique optimum $\{03, 05, 35\}$ (both checked exhaustively over all set partitions). In the notes' random data (computer evidence, $n \leq 9$) about 8–9% of instances force ≥ 2 edits on some packed P_3 and $\approx 0.2\%$ force three. For $c = 2$, $I(H)$ is subcubic but may have unbounded treewidth, so Theorem 6 is a different tradeoff from, not a recovery of, the $n^{2\ell}$ algorithm of [6].

Proposition 8 (Completing a fixed packed assignment)**.** *Let x assign a subset $x_i \subseteq \text{pairs}(P_i)$ to each P_i and let $B = |H| + \ell - \sum |x_i|$ be the excess budget left. Whether there is a set U of at most B unpacked pairs with $G \triangle (x \cup U)$ a cluster graph is decidable in $3^B n^{O(1)}$ time.*

Proof. Search tree on $G_x = G \triangle x$: if the current graph has an induced P_3, some pair of it must be flipped; packed pairs are frozen by x, so branch on its at most three unpacked pairs, reducing B by one (if all its pairs are packed, reject). Depth is at most B; flipping a pair twice is useless, so the search is exhaustive. □

Remark 9. Proposition 8 is a reformulation, not progress towards FPT in (ℓ, w): finding x is the whole problem, and the property "x is completable" is not local in x.

5 Limits of the parameterization

5.1 An ETH lower bound in the exponent

Theorem 10. *Assume ETH. There is no algorithm that decides CEaMP with $\ell = 0$ in time $2^{o(\text{tw}(I(H)))} n^{O(1)}$. This holds even when every vertex lies in at most 23 packed P_3's (so $I(H)$ has maximum degree at most 66), and whether or not a tree decomposition is supplied.*

Proof. Li, Pilipczuk and Sorge [6, Thm. 1] give a polynomial reduction from 3-SAT to CEaMP with $\ell = 0$ and at most 23 packed P_3's per vertex. We use it as a black box together with its size accounting. After a linear preprocessing every clause has exactly three distinct variables and every variable occurs at least twice; let N be the number of variables and m the number of clauses, so $\sum_i m_i = 3m$ for the occurrence counts m_i. By the gadget descriptions of [6] (read in the audit, not recomputed constant by constant): a variable gadget has $4m_i$ cliques of 5 vertices and 25 packed P_3's per triple of consecutive cliques, hence $20m_i$ vertices and $50m_i$ packed P_3's; a clause gadget has $20 + 114$ vertices (clique sizes $1, 1, 4, 14$ and $34, 46, 34$), $6 + 12$ transferring P_3's, and $O(1)$ padding P_3's from a triangle packing on cliques of constant size. Hence $|V| \approx 194m$ and $|H| = O(m)$ with an absolute constant (not computed; a crude count gives about $3000m$). By the Sparsification Lemma (Impagliazzo, Paturi and Zane; see also [3]) we may assume $m = O(N)$ at the cost of $2^{o(N)}$ extra time. Then $|H| = O(N)$ and $\text{tw}(I(H)) \leq |H| - 1 = O(N)$ (one bag). An algorithm with running time $2^{o(\text{tw}(I(H)))} n^{O(1)}$, which may compute its own decomposition, would therefore decide 3-SAT in time $2^{o(N)}$, contradicting ETH. □

Remark 11 (Caveats). (1) The bound is on the exponent only: it does not show that the base 3 (or 7) is tight, and $2^{o(w)}$ excludes $3^{o(w)}$ and $7^{o(w)}$ alike. A SETH-type bound $(3 - \varepsilon)^w$ is open. (2) The proof uses only $w \leq |H| - 1$; it does not exhibit hardness for instances of small $|H|$ relative to w. (3) The constants of the size table were read from the reduction and not re-derived one by one; the structural claim (every gadget has $O(1)$ size per clause or occurrence) was re-derived. (4) The statement refers to the version-3 arXiv text of [6].

5.2 Other parameters do not suffice

Proposition 12. *CEaMP is NP-hard when $I(H)$ has no edge ($w = 0$, vertex-disjoint packing), with ℓ part of the input. Hence w alone is not enough, and (by Theorem 1 of [6]) neither is $\ell = 0$ alone.*

Proof. Cluster Editing is NP-hard (e.g. already on graphs of maximum degree 6 [5]). Given (G, k) compute greedily a maximal set H of vertex-disjoint induced P_3's (if none, G is a cluster graph). Vertex-disjoint implies modification-disjoint and $I(H)$ is edgeless. Since $\text{OPT} \geq |H|$, (G, k) is a YES instance iff $(G, H, k - |H|)$ is (a NO instance if $k < |H|$). □

Proposition 13. $\mathrm{tw}(G)$ *and* $\mathrm{tw}(I(H))$ *are incomparable, in the following sense. (a) For the star* $G = K_{1,2m}$ *(*$m \geq 1$*) every packing* $H \neq \emptyset$ *has* $I(H) = K_{|H|}$ *and every YES instance* (G, H, ℓ) *satisfies* $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$*, while* $\mathrm{tw}(G) = 1$*. (b) For the* $k \times k$ *grid* G *and a family* H *of vertex-disjoint induced* P_3*'s along rows,* $\mathrm{tw}(I(H)) = 0$ *and* $\mathrm{tw}(G) = k$*.*

Proof. (a) Leaves have degree one, so the centre of any induced P_3 is the hub v; all packed P_3's are (a, v, b) with distinct leaves (the pairs va are disjoint), pairwise sharing v, so $I(H) = K_p$, $p = |H| \leq m$, $\mathrm{tw} = p - 1$. If the hub's cluster contains j leaves, the cost is $(2m - j) + \binom{j}{2}$ (the other leaves are best left as singletons), which is minimized, with value $2m - 1$, at $j \in \{1, 2\}$; so $\mathrm{OPT} = 2m - 1$ and a YES instance has $\ell \geq 2m - 1 - p$, i.e. $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$. (b) Rows of a grid are induced P_3's (no triangles) and $\mathrm{tw}(k \times k \text{ grid}) = k$ is standard. □

Thus hubs do not make $\mathrm{tw}(I(H))$ large by themselves; they force a tradeoff: small $\mathrm{tw}(I(H))$ on a star forces large ℓ.

6 Edits outside the packed P_3's

By Lemma 1 the whole difficulty of $\ell > 0$ is the set U of at most ℓ unpacked edits. We record what is proved about it.

Lemma 14 (Viability of an unpacked edit)**.** *Let* S *be any solution,* U *its unpacked part and* $uv \in U$*. Let* $D(uv)$ *be the set of* $z \notin Q(u) \cup Q(v) \cup \{u, v\}$ *such that* $\{u, v, z\}$ *is an induced* P_3 *of* $G \triangle \{uv\}$*. Then for every* $z \in D(uv)$ *one of* uz, vz *lies in* U*, and* $|D(uv)| \leq |U| - 1$*.*

Proof. All three pairs of $\{u, v, z\}$ are unpacked ($z \notin Q(u) \cup Q(v)$). If neither uz nor vz is in S, the triple has in $G \triangle S$ the same pattern as in $G \triangle \{uv\}$, an induced P_3. Distinct z give distinct pairs, all different from uv. □

Lemma 14 gives a kernel-style condition (deleting uv needs fewer than ℓ common neighbours outside $Q(u) \cup Q(v)$) but no bound on the number of candidate pairs: all pendant edges are viable. Moreover, a triple $\{u, v, z\}$ with uv unpacked that contains a packed pair has $z \in Q(u) \cup Q(v)$ (the packed pair is uz or vz), so an unpacked edit interacts with the P_3's only through the environments $Q(u), Q(v)$ of its endpoints.

Lemma 15 (Vertices outside $V(H)$ for maximal H)**.** *Let* H *be maximal and* $R = V \setminus V(H)$*. For* $r, s \in R$*: if* $rs \in E$ *then* $N[r] = N[s]$*; if* $rs \notin E$ *then* $N(r) \cap N(s) = \emptyset$*. Hence* R *is a union of true-twin cliques (components of* $G[R]$*) with pairwise disjoint neighbourhoods, and every neighbour of such a class outside it lies in* $V(H)$*.*

Proof. Packed pairs lie inside $V(H)$, so every pair at r or s is unpacked. If $rs \in E$ and $z \neq r, s$ is adjacent to exactly one of them, $\{r, s, z\}$ is an induced P_3 with all pairs unpacked. If $rs \notin E$ and z is a common neighbour, $r\,z\,s$ is one. Both contradict maximality. □

Proposition 16 (Strong locality fails)**.** *Let* $G = C_4$ *with edges* $01, 02, 13, 23$ *and* $H = \{(0, 1, 3)\}$ *(pairs* $01, 13, 03$*; vertex* $2 \notin V(H)$*). Then* H *is maximal,* $\mathrm{OPT} = 2 = |H| + 1$*, so* $(G, H, 0)$ *is a NO instance, and there are exactly three optima* $\{03, 12\}$*,* $\{01, 23\}$*,* $\{02, 13\}$*, each containing an unpacked pair at vertex* 2*.*

This (checked exhaustively over all partitions, and verified by hand: every induced P_3 of C_4 contains one of the packed pairs $01, 13, 03$) refutes only the claim that all unpacked edits of some optimum have both endpoints in $V(H)$; it does not refute weaker locality statements. In the notes' random sample ($n \leq 9$) about 4% of maximal-packing instances with $|R| \geq 2$ force an edit touching R (computer evidence, not a theorem).

7 Results pending verification

Everything in this section is unaudited. The statements come from the working notes (`wave2_c1_w1.md`, `wave2_hardness.md`); each has a proof sketch in the notes and computer checks on small instances ($n \le 11$), but none has passed the hostile-referee audit that the results of Sections 3–6 passed. They are labelled *Claim (unaudited)* so that they can be promoted to theorems after verification, and nothing else in the paper depends on them. Throughout, H is maximal and $R = V \setminus V(H)$.

Claim (unaudited) 1 (Twins). If $u \neq v$ and $N[u] = N[v]$, then u, v lie in one cluster in every optimal partition.

Sketch. Suppose $u \in C_u \neq C_v \ni v$. For $y \notin \{u, v\}$ and a set $D \not\ni u, v$ let $f(y, D)$ be the number of non-neighbours of y in D minus the number of neighbours; $f(u, D) = f(v, D)$. Moving u into C_v changes the cost by $D_1 = -f(u, C_u - u) + f(u, C_v - v) - 1$, moving v into C_u by $D_2 = -f(v, C_v - v) + f(v, C_u - u) - 1$; $D_1 + D_2 = -2 < 0$, so one move strictly improves. (Cf. the critical-clique property of Guo, TCS 410 (2009); not re-read.)

Claim (unaudited) 2 (C1: no unpacked edit inside R). Let $X_1, \dots, X_m$ be the components of $G[R]$ and $A_i = N(X_i) \setminus X_i \subseteq V(H)$ (pairwise disjoint, Lemma 15). Some optimal partition has every cluster containing no vertex of R or exactly one entire class X_i. In particular some optimum has no unpacked edit with both endpoints in R.

Sketch. By Claim 1 each X_i is inside one cluster, so an edit inside R exists only if a cluster C contains two classes X_1, X_2. Let $x_i = |X_i|$, $a_i = |A_i \cap C|$. Moving X_1 into a cluster of its own changes the cost by $\Delta_1 = x_1(x_1 + 2a_1 - |C|)$, and $|C| \ge x_1 + x_2 + a_1 + a_2$. If $\Delta_1, \Delta_2 > 0$ then $x_2 + a_2 < a_1$ and $x_1 + a_1 < a_2$, so $x_1 + x_2 < 0$, absurd. So one class can be moved out for free; iterate.

Claim (unaudited) 3 (Theorem K: FPT in (ℓ, s)). Let s be the maximum number of vertices of a connected component of $I(H)$ (the union of the vertex sets of its P_3's). Then CEaMP is FPT in (ℓ, s); in particular it is FPT when $I(H)$ is a perfect matching.

Sketch. Components have disjoint vertex sets W_j. Compute $\tau_j = \mathrm{OPT}(G[W_j]) \ge |C_j|$ by brute force (exponential in s only). Then "$\mathrm{OPT}(G) \le |H| + \ell$" is Cluster Editing with $k = |H| + \ell$ and a vertex-disjoint packing $\{G[W_j]\}$ of cost $t \le \binom{s}{2}$, with excess $\ell' = \ell - \sum_j (\tau_j - |C_j|)$ (NO if negative); apply the above-lower-bound algorithm of van Bevern et al. [9, Thm. 6.6, Cor. 6.7] (time $f(t, \ell')n^{O(1)}$). A pair with exactly one end in W_j is never packed, which restores the property of [9] that external edits are excess. *Depends on a cited theorem used outside its stated setting, to be checked.* Not valid for trees of unbounded size.

Claim (unaudited) 4 (MAX: maximalisation costs $+\ell$ in treewidth). Let M be a maximal set of induced P_3's with pairwise disjoint pair sets, disjoint from the pairs of H. Then $\mathrm{OPT} \le |H| + \ell$ iff $|M| \le \ell$ and $\mathrm{OPT} \le |H \cup M| + (\ell - |M|)$, and $\mathrm{tw}(I(H \cup M)) \le w + |M| \le w + \ell$.

Sketch. Remark 2 for the first part; adding $|M|$ nodes to $I(H)$ and putting them in every bag raises the width by at most $|M|$.

Claim (unaudited) 5 (HUB). Let D be the maximum number of packed partners of a vertex ($D \le 2(w+1)$). If u has non-packed neighbours X that are pairwise non-adjacent with no packed pair among them, then $\lfloor |X|/2 \rfloor \le \mathrm{OPT} - |H|$. Hence $\alpha(G[N_h(u)]) \le (D+1)(2\ell+1)$ in a YES instance, where $N_h(u)$ are the non-packed neighbours.

Sketch. $(x_1, u, x_2), (x_3, u, x_4), \dots$ are induced P_3's with all pairs unpacked, pair-disjoint and disjoint from H, so $\mathrm{OPT} \ge |H| + \lfloor |X|/2 \rfloor$. Each x has at most D packed partners, so an independent set of size q contains such an X with $|X| \ge q/(D+1)$.

Claim (unaudited) 6 (SM: small merge). Let S be a solution with $|U| \le \ell$, D_s the maximum packed degree, and let the *pieces* be the connected components of $(V, E \setminus \text{packed} \setminus F)$, F the deleted unpacked edges. If two distinct pieces A, B lie in one cluster, then every pair ab ($a \in A$, $b \in B$) is packed or an inserted unpacked pair, hence $\min(|A|,|B|)(\max(|A|,|B|) - D_s) \le |U|$ and $\max(|A|,|B|) \le D_s + |U|$; a cluster with at least two pieces has at most $\max(2D + \ell, 2\ell)$ vertices.

Sketch. A hard edge between a and b not in F would put them in one piece; so ab is packed or inserted. There are at most $D_s \min(|A|,|B|)$ packed cross pairs and at most $|U|$ inserted ones; summing over piece pairs gives the cluster bound. Consequence in the notes: the natural Multicolored-Clique encodings of hub/tile/slip-chain type collapse, which is why no W[1]-hardness reduction was found (evidence, not a proof of FPT).

8 Open problems

Notation as above; $w = \text{tw}(I(H))$.

Open Problem 1 (Main question)**.** Is CEaMP fixed-parameter tractable parameterized by $\ell + w$, i.e. solvable in $f(\ell, w)n^{O(1)}$ time, or is it W[1]-hard in $\ell + w$? Known: XP in ℓ and FPT in w for each fixed ℓ only via $n^{2\ell}$ (Theorem 6); FPT for $w = 0$ by [9]; the case $w = 0$ and the case $H = \emptyset$ (plain Cluster Editing, $\ell = k$) are both FPT. Note that adding c to the parameter changes nothing, since $c \le w + 1$. A positive answer needs a bound $g(\ell, w)$ on the number of candidate positions for each unpacked edit (Lemma 14 gives none); a negative answer has to encode $[n]$-valued information through unpacked edits.

Open Problem 2 (Forests)**.** If $I(H)$ is a forest ($w \le 1$, hence $c \le 2$), is CEaMP FPT in ℓ? (Perfect-matching $I(H)$ is covered only by the unaudited Claim 3.)

Open Problem 3 (Half-integral and intermediate packings)**.** Is CEaMP with $c = 2$ FPT in ℓ? What is the complexity of $\ell = 0$ for $3 \le c \le 22$? Both are open in [6], whose authors say a W[1]-hardness result for $c = 2$ "would also not be surprising".

Open Problem 4 (Base of the exponent)**.** For $\ell = 0$, does a $(3 - \varepsilon)^w n^{O(1)}$ (or even $2^{o(w)}$ with a smaller base) algorithm exist, or is there a SETH-type lower bound? Theorem 10 gives only the exponent.

Open Problem 5 (Treewidth of G)**.** What is the complexity of CEaMP with $\ell = 0$ (or of Cluster Editing) when $\text{tw}(G)$ is bounded? We found no published result on Cluster Editing parameterized by treewidth alone (Section 9), so this must be proved rather than cited.

9 Related work

Packing lower bounds. van Bevern, Froese and Komusiewicz [9] parameterize F-free editing above the size of a vertex-disjoint packing; for Cluster Editing they obtain an $O(4^\ell \ell^3 + m + n)$-time algorithm for P_3 packings (Thm. 6.13), and for cost-t packings an $O(1.62^{(2t+1)\ell} + nm + n1.62^t)$ algorithm and a kernel with $O(t\ell)$ vertices. They show NP-hardness at $\ell = 0$ of K_3-free editing for edge-disjoint packings and of K_q-free editing, $q \ge 6$, for vertex-disjoint packings, and leave open Cluster Editing above edge-disjoint packings and the stronger modification-disjoint bound. Li, Pilipczuk and Sorge [6] answer the latter: NP-hard at $\ell = 0$ with at most 23 packed P_3's per vertex, $n^{2\ell+O(1)}$ for $c = 2$, and open for $c = 2$ in ℓ and for $3 \le c \le 22$ at $\ell = 0$. They report "grid-like structures" as the obstacle. Our $\text{tw}(I(H))$ parameter does not appear in either paper.

Citations and treewidth. Of the citing works recorded by the notes, we opened the survey of Crespelle et al. [2], the vertex-splitting paper [8] (it uses modification-disjoint P_3 packings as a lower-bound tool), Bocci et al. [1] and Garvardt et al. [4]; none improves $c = 2$, settles $3 \leq c \leq 22$, or uses a treewidth-type parameter. Two further citing items (the PACE 2021 report and a 2021 thesis) were not opened, and citation coverage was incomplete (Semantic Scholar listed seven records, OpenAlex a count of two); this is a "not found", not proof of absence. We found no paper stating the complexity of plain Cluster Editing or Deletion parameterized by treewidth. The nearby s-Club Cluster Edge Deletion is FPT in $s + \mathrm{tw}$ but W[1]-hard in pathwidth [7]; as a deletion problem with a diameter bound it does not transfer to editing. Komusiewicz and Uhlmann [5] show that Cluster Editing is NP-hard at maximum degree 6, with no $2^{o(k)}$ algorithm under ETH even at bounded degree; this is consistent with the fact that bounded degree of $I(H)$ does not give tractability (Theorem 10).

References

[1] Bocci et al. A new temporal interpretation of cluster editing, 2022. Notes: opened. TODO: full author list.

[2] Crespelle, Drange, Fomin, and Golovach. A survey of parameterized algorithms and the complexity of edge modification, 2020. Notes: opened.

[3] Cygan et al. *Parameterized Algorithms*. Springer, 2015. Notes (audit.md): recommended reference for dynamic programming on tree decompositions and for ETH lower bounds; the specific theorems were not opened.

[4] Garvardt, Morawietz, Nichterlein, and Weller. Graph clustering problems under the lens of parameterized local search. In *Proc. IPEC 2023*, volume 285 of *LIPIcs*, page Art. 20, 2023. Notes: opened (Dagstuhl PDF); the DOI is read off the recorded Dagstuhl URL. Journal version JCSS 2026 per the notes.

[5] Komusiewicz and Uhlmann. Cluster editing with locally bounded modifications. *Discrete Applied Mathematics*, 2012. Notes: opened, `https://www.uni-marburg.de/de/fb12/arbeitsgruppen/algorith/paper/ce-bounded-degree.pdf`. Volume and pages not recorded in the notes.

[6] Li, Pilipczuk, and Sorge. Cluster editing parameterized above modification-disjoint P_3-packings. *ACM Transactions on Algorithms*, 20(1):Art. 3, 2023. Conference version: STACS 2021, LIPIcs 187, Art. 49. Notes: opened (arXiv v3). Surnames only, as recorded in the notes.

[7] Montecchiani et al. s-club cluster edge deletion, 2025. Notes: opened. Title and author list to be checked.

[8] Authors not recorded in the notes. The complexity of cluster vertex splitting and company, 2023. Notes: opened. TODO: fill in authors from the arXiv page.

[9] van Bevern, Froese, and Komusiewicz. Parameterizing edge modification problems above lower bounds. *Theory of Computing Systems*, 62(3), 2018. Conference version: CSR 2016. Notes: opened (arXiv).